

```
In [8]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [9]: import requests
```

```
In [10]: response = requests.get("https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv")
```

```
In [11]: data = response.json
data
```

```
Out[11]: <bound method Response.json of <Response [200]>>
```

```
In [28]: #df = pd.DataFrame(data)
#print(df.head())
```

```
In [13]: from io import StringIO

# Step 1: Get the CSV data as text
#response = requests.get("https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv")

# Step 2: Convert to a pandas DataFrame using StringIO
if response.status_code == 200:
    csv_text = response.text
    data = StringIO(csv_text)

    # Specify column names since the dataset has no header row
    column_names = [
        'Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness',
        'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'
    ]

    df = pd.read_csv(data, names=column_names)
    print(df.head())
else:
    print("Failed to fetch CSV:", response.status_code)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
In [14]: df.columns
```

```
Out[14]: Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
               'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
              dtype='object')
```

```
In [15]: df.shape
```

```
Out[15]: (768, 9)
```

```
In [16]: df.count
```

```
Out[16]: <bound method DataFrame.count of
0      6    148    72    35    0  33.6
1      1     85    66    29    0  26.6
2      8    183    64     0    0  23.3
3      1     89    66    23    94  28.1
4      0    137    40    35   168  43.1
...    ...    ...    ...    ...    ...
763    10    101    76    48   180  32.9
764     2    122    70    27    0  36.8
765     5    121    72    23   112  26.2
766     1    126    60     0    0  30.1
767     1     93    70    31    0  30.4

DiabetesPedigreeFunction  Age  Outcome
0      0.627    50      1
1      0.351    31      0
2      0.672    32      1
3      0.167    21      0
4      2.288    33      1
..    ...    ...    ...
763    0.171    63      0
764    0.340    27      0
765    0.245    30      0
766    0.349    47      1
767    0.315    23      0

[768 rows x 9 columns]>
```

```
In [17]: df.isnull().sum()
```

```
Out[17]: Pregnancies      0
Glucose      0
BloodPressure  0
SkinThickness  0
Insulin      0
BMI          0
DiabetesPedigreeFunction  0
Age          0
Outcome      0
dtype: int64
```

Split into features (X) and target (y)

```
In [20]: from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

```
In [36]: #pip install optuna
```

```
Collecting optuna
  Downloading optuna-4.3.0-py3-none-any.whl.metadata (17 kB)
Collecting alembic<=1.5.0 (from optuna)
  Downloading alembic-1.16.1-py3-none-any.whl.metadata (7.3 kB)
Collecting colorlog (from optuna)
  Downloading colorlog-6.9.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: numpy in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from optuna) (1.26.4)
Requirement already satisfied: packaging>=20.0 in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from optuna) (24.1)
Requirement already satisfied: sqlalchemy>=1.4.2 in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from optuna) (2.0.36)
Requirement already satisfied: tqdm in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from optuna) (4.66.5)
Requirement already satisfied: PyYAML in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from optuna) (6.0.2)
Collecting Mako (from alembic<=1.5.0->optuna)
  Downloading mako-1.3.10-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: typing-extensions>=4.12 in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from alembic<=1.5.0->optuna) (4.12.2)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from sqlalchemy>=1.4.2->optuna) (3.1.1)
Requirement already satisfied: colorama in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from colorlog->optuna) (0.4.6)
Requirement already satisfied: MarkupSafe>=0.9.2 in c:\users\shaw3\appdata\local\programs\python\python312\lib\site-packages (from Mako->alembic<=1.5.0->optuna) (2.1.5)
Downloading optuna-4.3.0-py3-none-any.whl (386 kB)
Downloading alembic-1.16.1-py3-none-any.whl (242 kB)
Downloading colorlog-6.9.0-py3-none-any.whl (11 kB)
Downloading mako-1.3.10-py3-none-any.whl (78 kB)
Installing collected packages: Mako, colorlog, alembic, optuna
Successfully installed Mako-1.3.10 alembic-1.16.1 colorlog-6.9.0 optuna-4.3.0
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 24.2 -> 25.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [37]: import optuna
```

```
In [38]: X= df.drop('Outcome', axis=1)
y = df['Outcome']

X_train, X_test, y_train, y_test=train_test_split(X,y, test_size= 0.3, random_state= 42)
```

```
In [25]: # Optional: Scale the data for better model performance
scaler= StandardScaler()
X_train= scaler.fit_transform(X_train)
X_test= scaler.transform(X_test)
```

```
In [26]: # Check the shape of the data
print(f'Training set shape: {X_train.shape}')
print(f'Test set shape: {X_test.shape}')
```

```
Training set shape: (537, 8)
Test set shape: (231, 8)
```

```
In [27]: from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score
```

```
In [64]: # Define the objective function
def objective(trial):
    # Suggest values for the hyperparameters
    n_estimators =trial.suggest_int('n_estimators', 50, 200)
    max_depth= trial.suggest_int('max_depth', 3, 20)

    # Create the RandomForestClassifier with suggested hyperparameters
    model= RandomForestClassifier(
        n_estimators=n_estimators,
        max_depth= max_depth,
        random_state= 42
    )
    # Perform 3-fold cross-validation and calculate accuracy
    score= cross_val_score(model, X_train, y_train, cv=3,scoring='accuracy').mean()
    return score # Return the accuracy score for Optuna to maximize
```

Each trial represents one attempt to find a good set of hyperparameters

n_estimators → number of trees in the forest (50 to 200)

max_depth → maximum depth of each tree (3 to 20)

.mean() returns the average accuracy over all 3 folds.

.Splits your training data (X_train, y_train) into 3 parts.

Trains the model on 2 parts and validates on the 3rd — 3 times.

```
In [65]: # Create a study object and optimize the objective function
study = optuna.create_study(direction='maximize', sampler=optuna.samplers.TPESampler()) # We aim to maximize the accuracy
study.optimize(objective, n_trials=50) # we Run 50 trials to find the best hyperparameters
```

[I 2025-06-12 22:35:43,867] A new study created in memory with name: no-name-ffb9acbc-2694-416f-bdc6-b4a59f15c284
[I 2025-06-12 22:35:44,484] Trial 0 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 120, 'max_depth': 14}. Best is trial 0 with value: 0.7746741154562384.
[I 2025-06-12 22:35:45,134] Trial 1 finished with value: 0.7579143389199254 and parameters: {'n_estimators': 162, 'max_depth': 17}. Best is trial 0 with value: 0.7746741154562384.
[I 2025-06-12 22:35:45,371] Trial 2 finished with value: 0.7914338919925511 and parameters: {'n_estimators': 59, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:45,623] Trial 3 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 73, 'max_depth': 3}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:46,311] Trial 4 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 165, 'max_depth': 19}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:46,634] Trial 5 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 83, 'max_depth': 12}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:47,173] Trial 6 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 142, 'max_depth': 6}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:47,890] Trial 7 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 190, 'max_depth': 13}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:48,579] Trial 8 finished with value: 0.7802607076350093 and parameters: {'n_estimators': 180, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:49,200] Trial 9 finished with value: 0.7821229050279329 and parameters: {'n_estimators': 163, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:49,409] Trial 10 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 55, 'max_depth': 3}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:49,844] Trial 11 finished with value: 0.7895716945996276 and parameters: {'n_estimators': 114, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:50,263] Trial 12 finished with value: 0.7839851024208566 and parameters: {'n_estimators': 107, 'max_depth': 9}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:50,638] Trial 13 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 97, 'max_depth': 10}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:50,851] Trial 14 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 55, 'max_depth': 5}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:51,334] Trial 15 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 134, 'max_depth': 6}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:51,684] Trial 16 finished with value: 0.7765363128491621 and parameters: {'n_estimators': 89, 'max_depth': 15}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:52,139] Trial 17 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 115, 'max_depth': 11}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:52,422] Trial 18 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 71, 'max_depth': 7}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:52,931] Trial 19 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 132, 'max_depth': 11}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:53,130] Trial 20 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 50, 'max_depth': 5}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:53,559] Trial 21 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 105, 'max_depth': 9}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:54,188] Trial 22 finished with value: 0.7821229050279329 and parameters: {'n_estimators': 147, 'max_depth': 9}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:54,612] Trial 23 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 110, 'max_depth': 10}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:54,903] Trial 24 finished with value: 0.7877094972067039 and parameters: {'n_estimators': 71, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:55,240] Trial 25 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 84, 'max_depth': 7}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:55,489] Trial 26 finished with value: 0.7579143389199254 and parameters: {'n_estimators': 68, 'max_depth': 5}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:55,757] Trial 27 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 66, 'max_depth': 7}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:56,131] Trial 28 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 95, 'max_depth': 12}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:56,465] Trial 29 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 79, 'max_depth': 16}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:56,949] Trial 30 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 120, 'max_depth': 13}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:57,366] Trial 31 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 105, 'max_depth': 9}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:57,629] Trial 32 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 63, 'max_depth': 10}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:58,124] Trial 33 finished with value: 0.7895716945996275 and parameters: {'n_estimators': 127, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:58,615] Trial 34 finished with value: 0.7895716945996276 and parameters: {'n_estimators': 126, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:59,060] Trial 35 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 127, 'max_depth': 4}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:35:59,641] Trial 36 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 152, 'max_depth': 6}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:00,183] Trial 37 finished with value: 0.7579143389199254 and parameters: {'n_estimators': 139, 'max_depth': 20}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:00,782] Trial 38 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 155, 'max_depth': 7}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:01,438] Trial 39 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 171, 'max_depth': 18}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:01,892] Trial 40 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 125, 'max_depth': 4}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:02,343] Trial 41 finished with value: 0.7839851024208566 and parameters: {'n_estimators': 119, 'max_de

```

pth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:02,654] Trial 42 finished with value: 0.7877094972067038 and parameters: {'n_estimators': 79, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:03,055] Trial 43 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 97, 'max_depth': 10}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:03,296] Trial 44 finished with value: 0.7914338919925511 and parameters: {'n_estimators': 59, 'max_depth': 8}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:03,823] Trial 45 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 142, 'max_depth': 6}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:04,654] Trial 46 finished with value: 0.7821229050279329 and parameters: {'n_estimators': 198, 'max_depth': 9}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:04,911] Trial 47 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 59, 'max_depth': 11}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:05,746] Trial 48 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 172, 'max_depth': 13}. Best is trial 2 with value: 0.7914338919925511.
[I 2025-06-12 22:36:06,224] Trial 49 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 114, 'max_depth': 7}. Best is trial 2 with value: 0.7914338919925511.

```

```

In [66]: # Print the best result
print(f'Best trial accuracy: {study.best_trial.value}')
print(f'Best hyperparameters: {study.best_trial.params}')

```

```

Best trial accuracy: 0.7914338919925511
Best hyperparameters: {'n_estimators': 59, 'max_depth': 8}

```

```

In [ ]:

```

```

In [48]: from sklearn.metrics import accuracy_score

# Train a RandomForestClassifier using the best hyperparameters from Optuna
best_model= RandomForestClassifier(**study.best_trial.params, random_state= 42)

X_pred= best_model.fit(X_train, y_train)

y_pred= best_model.predict(X_test)

# Calculate the accuracy on the test set
test_accuracy = accuracy_score(y_test, y_pred)

# Print the test accuracy
print(f'Test accuracy with the best hyperparameters: {test_accuracy: .2f}')

```

```

Test accuracy with the best hyperparameters: 0.74

```

```

In [49]: search_space= {
    'n_estimators': [50, 100, 150, 200],
    'max_depth': [5, 10, 15, 20]
}

```

This search space is used in manual search or grid/random search-based tuning methods. we're telling your tuning algorithm:

“Try combinations of these values and find the best one.”

```

In [ ]:

```

Create a study and optimize it using GridSampler

```

In [53]: study= optuna.create_study(direction='maximize', sampler= optuna.samplers.GridSampler(search_space))
study.optimize(objective)

```

```
[I 2025-06-12 22:26:12,823] A new study created in memory with name: no-name-064b3f45-762f-4b01-801a-fa22e43f5f0a
[I 2025-06-12 22:26:13,337] Trial 0 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 100, 'max_depth': 5}. Best is trial 0 with value: 0.7709497206703911.
[I 2025-06-12 22:26:13,936] Trial 1 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 150, 'max_depth': 10}. Best is trial 0 with value: 0.7709497206703911.
[I 2025-06-12 22:26:14,221] Trial 2 finished with value: 0.7839851024208566 and parameters: {'n_estimators': 50, 'max_depth': 15}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:14,620] Trial 3 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 100, 'max_depth': 15}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:15,001] Trial 4 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 100, 'max_depth': 20}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:15,201] Trial 5 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 50, 'max_depth': 10}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:15,748] Trial 6 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 150, 'max_depth': 5}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:16,332] Trial 7 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 150, 'max_depth': 20}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:16,897] Trial 8 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 150, 'max_depth': 15}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:17,620] Trial 9 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 200, 'max_depth': 10}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:18,379] Trial 10 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 200, 'max_depth': 20}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:19,107] Trial 11 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 200, 'max_depth': 15}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:19,812] Trial 12 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 200, 'max_depth': 5}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:20,008] Trial 13 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 50, 'max_depth': 5}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:20,375] Trial 14 finished with value: 0.7597765363128492 and parameters: {'n_estimators': 100, 'max_depth': 10}. Best is trial 2 with value: 0.7839851024208566.
[I 2025-06-12 22:26:20,578] Trial 15 finished with value: 0.7802607076350093 and parameters: {'n_estimators': 50, 'max_depth': 20}. Best is trial 2 with value: 0.7839851024208566.
```

This code uses Optuna with a GridSampler to systematically try all combinations of specified hyperparameters (like `n_estimators` and `max_depth`) to find the best model. The `create_study` function initializes the optimization process with the goal of maximizing a metric (like accuracy), and the `study.optimize(objective)` runs the model training and evaluation for each combination. Unlike random or adaptive sampling, GridSampler ensures that every possible combination in the search space is tested, making it a thorough but potentially time-consuming method.

Print the best result

```
In [59]: print(f'Best trial accuracy: {study.best_trial.value}')
         print(f'Best hyperparameters: {study.best_trial.params}')

Best trial accuracy: 0.7839851024208566
Best hyperparameters: {'n_estimators': 50, 'max_depth': 15}
```

classification report check

```
In [58]: from sklearn.metrics import classification_report

         print(classification_report(y_test, y_pred))
```

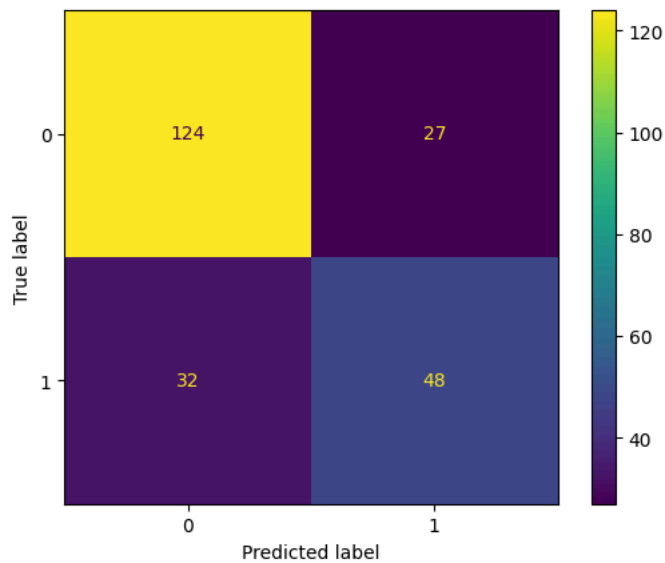
	precision	recall	f1-score	support
0	0.79	0.82	0.81	151
1	0.64	0.60	0.62	80
accuracy			0.74	231
macro avg	0.72	0.71	0.71	231
weighted avg	0.74	0.74	0.74	231

```
In [69]: from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

         y_pred = best_model.predict(X_test)
```

```
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
```

Out[69]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x1a0b5219430>



ROC-AUC Score (for binary classification)

```
In [70]: from sklearn.metrics import roc_auc_score

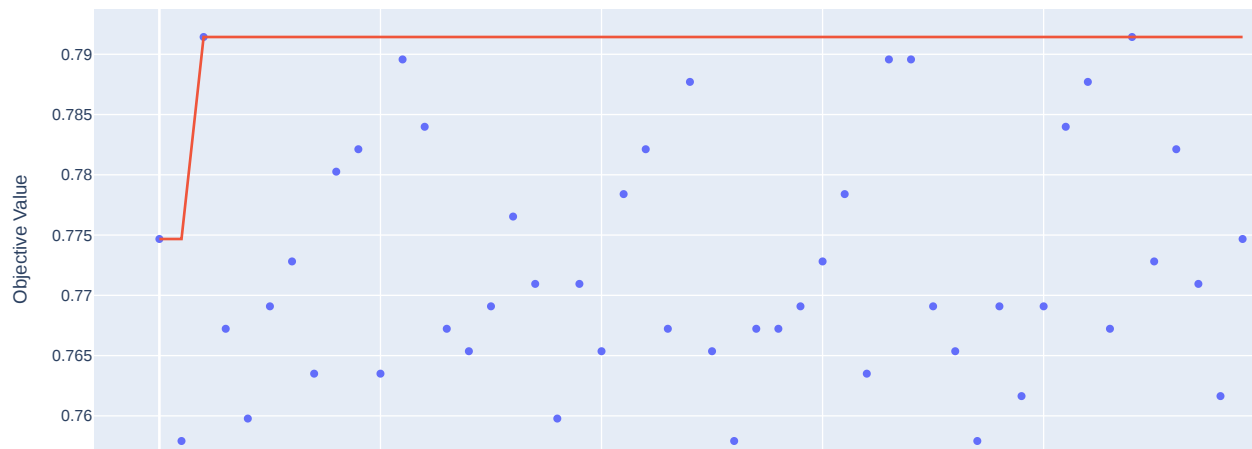
y_proba = best_model.predict_proba(X_test)[: , 1]
roc_score = roc_auc_score(y_test, y_proba)
print("ROC-AUC Score:", roc_score)
```

ROC-AUC Score: 0.8021523178807947

Optuna Visualizations

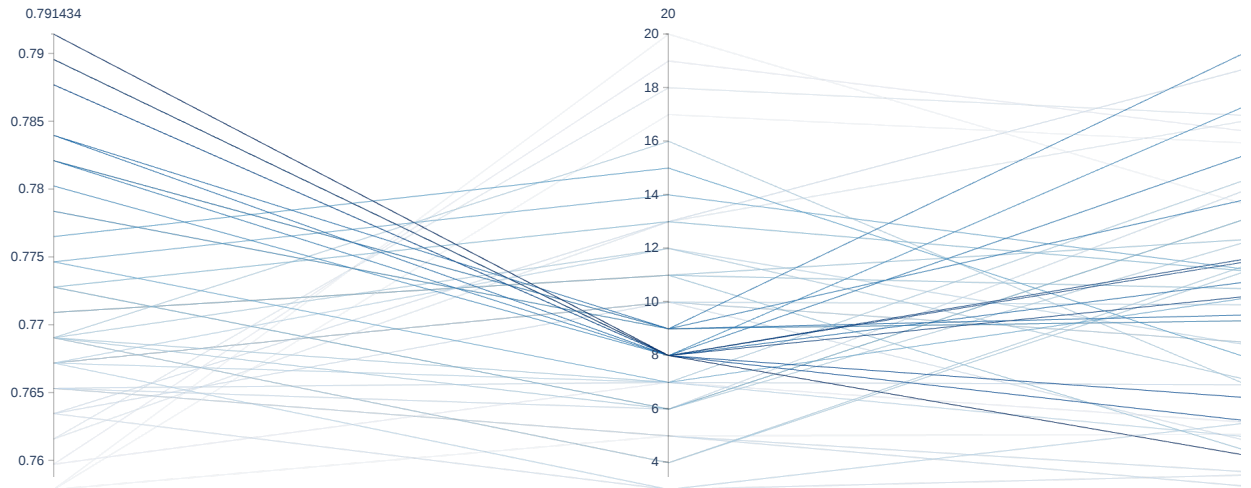
```
In [73]: from optuna.visualization import plot_optimization_history, plot_parallel_coordinate, plot_slice, plot_contour, plot_parallel_coordinate
plot_optimization_history(study).show()
```

Optimization History Plot



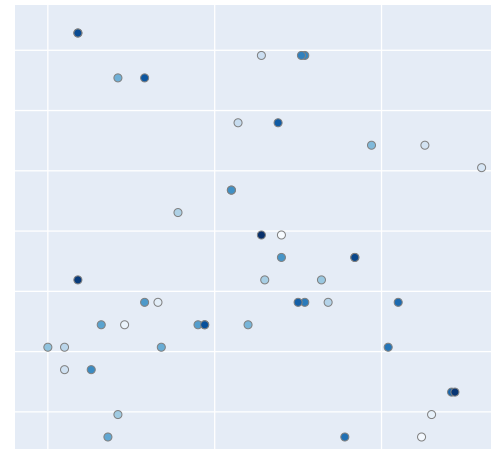
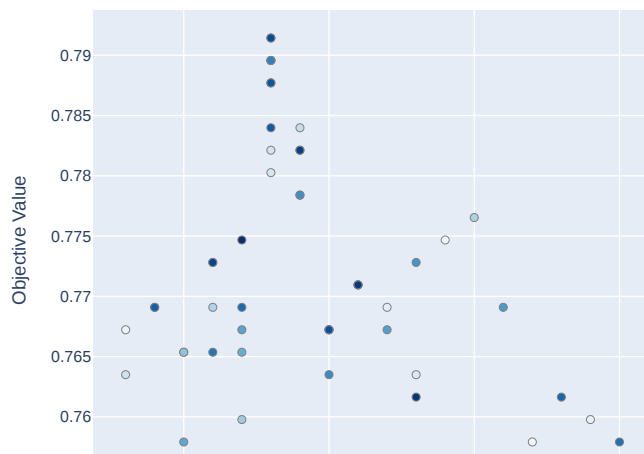
```
In [75]: from optuna.visualization import plot_parallel_coordinate
plot_parallel_coordinate(study).show()
```

Parallel Coordinate Plot



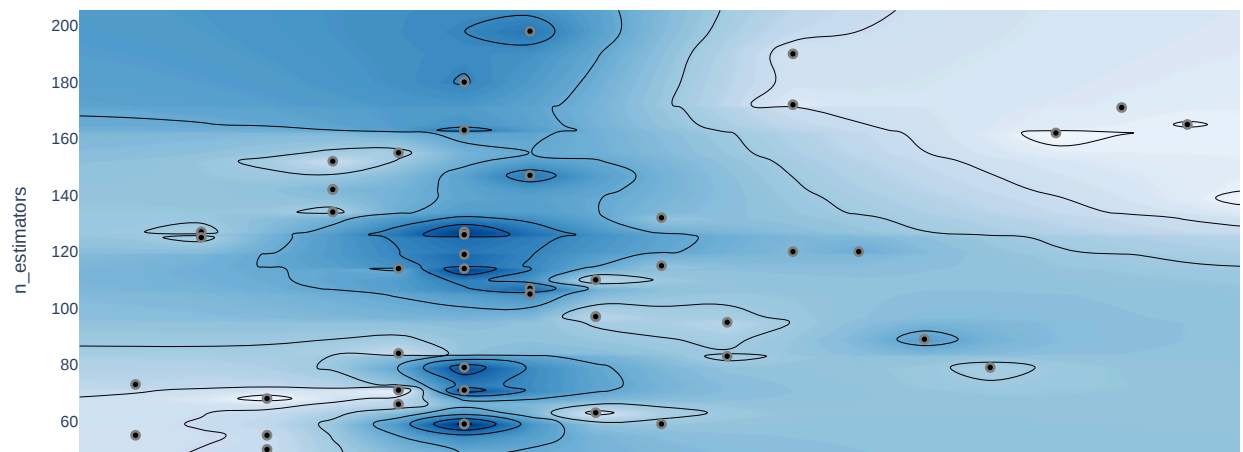
```
In [76]: from optuna.visualization import plot_slice
plot_slice(study).show()
```

Slice Plot



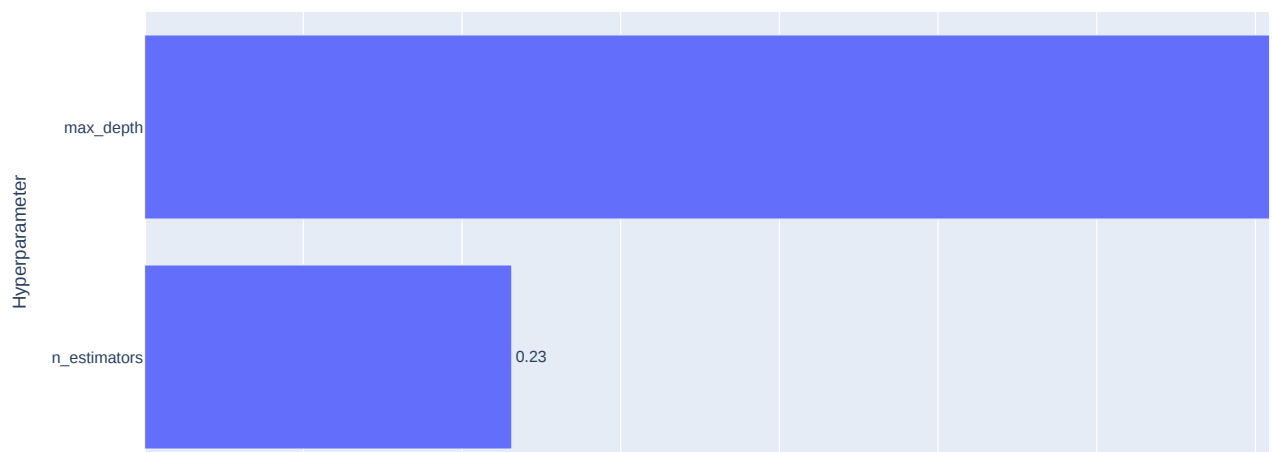
```
In [77]: from optuna.visualization import plot_contour
plot_contour(study).show()
```


Contour Plot



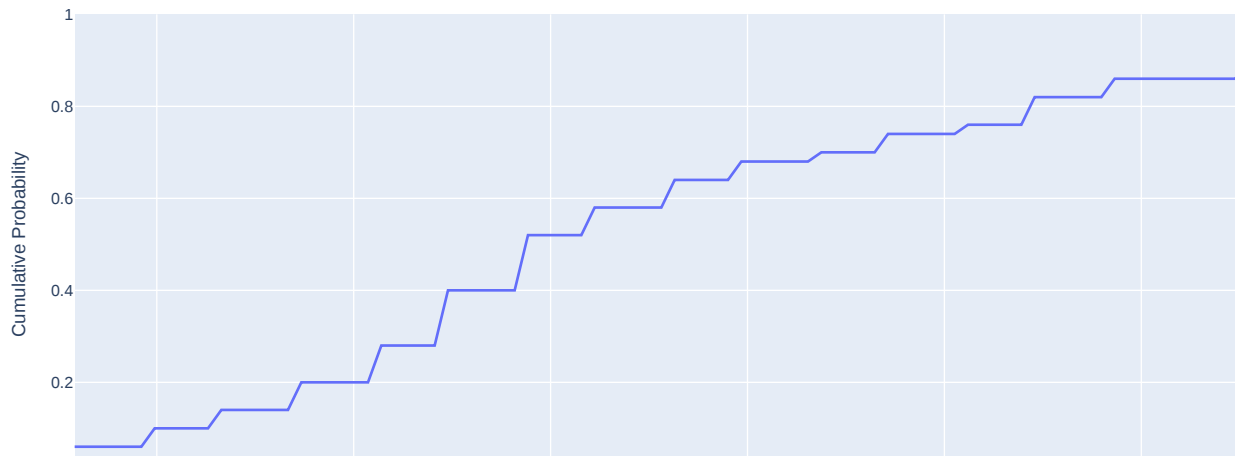
```
In [78]: from optuna.visualization import plot_param_importances
plot_param_importances(study).show()
```

Hyperparameter Importances



```
In [80]: from optuna.visualization import plot_edf
plot_edf(study).show()
```

Empirical Distribution Function Plot



Optimizing Multiple ML Models

```
In [81]: from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.svm import SVC
```

```
In [94]: from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.metrics import accuracy_score

def objective(trial):
    # Choose the algorithm to tune
    classifier_name = trial.suggest_categorical('classifier', ['SVM', 'RandomForest', 'GradientBoosting'])

    if classifier_name == 'SVM':
        # SVM hyperparameters
        C = trial.suggest_float('C', 0.1, 100, log=True)
        kernel = trial.suggest_categorical('kernel', ['linear', 'poly', 'sigmoid', 'rbf'])
        gamma = trial.suggest_categorical('gamma', ['scale', 'auto'])

        model = SVC(C=C, gamma=gamma, kernel=kernel, random_state=42)

    elif classifier_name == 'RandomForest':
        # Random Forest hyperparameters
        n_estimators = trial.suggest_int('n_estimators', 50, 300)
        max_depth = trial.suggest_int('max_depth', 3, 20)
        min_samples_split = trial.suggest_int('min_samples_split', 2, 10)
        min_samples_leaf = trial.suggest_int('min_samples_leaf', 1, 10)
        bootstrap = trial.suggest_categorical('bootstrap', [True, False])

        model = RandomForestClassifier(
            n_estimators=n_estimators,
            max_depth=max_depth,
            min_samples_split=min_samples_split,
            min_samples_leaf=min_samples_leaf,
            bootstrap=bootstrap,
            random_state=42
        )

    elif classifier_name == 'GradientBoosting':
        # Gradient Boosting hyperparameters
        n_estimators = trial.suggest_int('n_estimators', 50, 300)
        learning_rate = trial.suggest_float('learning_rate', 0.01, 0.3, log=True)
        max_depth = trial.suggest_int('max_depth', 3, 20)
        min_samples_split = trial.suggest_int('min_samples_split', 2, 10)
        min_samples_leaf = trial.suggest_int('min_samples_leaf', 1, 10)

        model = GradientBoostingClassifier(
            n_estimators=n_estimators,
            learning_rate=learning_rate,
            max_depth=max_depth,
            min_samples_split=min_samples_split,
```

```

        min_samples_leaf=min_samples_leaf,
        random_state=42
    )

    # Perform cross-validation and return the mean accuracy
    score = cross_val_score(model, X_train, y_train, cv=3, scoring='accuracy').mean()
    return score

```

C is a regularization parameter that controls how much the model tries to avoid misclassifying each training example.

Higher C = less regularization (fit the training data better, but risk overfitting).

log=True means Optuna will search values like 0.1, 1, 10, 100 — spread on a logarithmic scale.

'linear' for straight lines

'rbf' for curved, flexible decision boundaries

gamma controls how far the influence of a single training point reaches.

'scale' and 'auto' are predefined methods for calculating it.

In []:

random_state=42 is for reproducibility — so results are consistent across runs.

In []:

2nd part of the code, we define how Optuna should tune a Random Forest Classifier by searching for the best combination of its key hyperparameters. The function allows Optuna to try different values for the number of trees (n_estimators), the maximum depth of each tree (max_depth), the minimum number of samples needed to split a node (min_samples_split), the minimum number of samples required at each leaf (min_samples_leaf), and whether bootstrapping is used (bootstrap). These hyperparameters control how the Random Forest learns from data—affecting both accuracy and generalization.

How deep each tree can grow.

A deeper tree captures more complexity but may overfit.

```

In [95]: # Create a study and optimize it using CmaEsSampler
study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=100)

```

[I 2025-06-12 23:27:39,556] A new study created in memory with name: no-name-338dcbcf-1710-413f-9a49-95d87f8def59
[I 2025-06-12 23:27:40,986] Trial 0 finished with value: 0.7597765363128491 and parameters: {'classifier': 'RandomForest', 'n_estimators': 251, 'max_depth': 4, 'min_samples_split': 6, 'min_samples_leaf': 4, 'bootstrap': True}. Best is trial 0 with value: 0.7597765363128491.
[I 2025-06-12 23:27:43,782] Trial 1 finished with value: 0.7467411545623835 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 159, 'learning_rate': 0.11583426840081856, 'max_depth': 16, 'min_samples_split': 8, 'min_samples_leaf': 1}. Best is trial 0 with value: 0.7597765363128491.
[I 2025-06-12 23:27:45,236] Trial 2 finished with value: 0.7728119180633147 and parameters: {'classifier': 'SVM', 'C': 0.37105269891878623, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 2 with value: 0.7728119180633147.
[I 2025-06-12 23:27:46,073] Trial 3 finished with value: 0.7690875232774674 and parameters: {'classifier': 'RandomForest', 'n_estimators': 290, 'max_depth': 20, 'min_samples_split': 7, 'min_samples_leaf': 10, 'bootstrap': False}. Best is trial 2 with value: 0.7728119180633147.
[I 2025-06-12 23:27:47,848] Trial 4 finished with value: 0.7411545623836128 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 228, 'learning_rate': 0.2674902286618688, 'max_depth': 20, 'min_samples_split': 7, 'min_samples_leaf': 10}. Best is trial 2 with value: 0.7728119180633147.
[I 2025-06-12 23:27:47,870] Trial 5 finished with value: 0.6499068901303539 and parameters: {'classifier': 'SVM', 'C': 25.28400259639163, 'kernel': 'sigmoid', 'gamma': 'auto'}. Best is trial 2 with value: 0.7728119180633147.
[I 2025-06-12 23:27:50,053] Trial 6 finished with value: 0.7486033519553073 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 272, 'learning_rate': 0.05800730667788165, 'max_depth': 10, 'min_samples_split': 5, 'min_samples_leaf': 8}. Best is trial 2 with value: 0.7728119180633147.
[I 2025-06-12 23:27:50,571] Trial 7 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 147, 'max_depth': 9, 'min_samples_split': 4, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:50,634] Trial 8 finished with value: 0.6499068901303539 and parameters: {'classifier': 'SVM', 'C': 7.760060986029223, 'kernel': 'rbf', 'gamma': 'auto'}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:52,384] Trial 9 finished with value: 0.7728119180633147 and parameters: {'classifier': 'SVM', 'C': 0.6121155629468422, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:52,626] Trial 10 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 60, 'max_depth': 7, 'min_samples_split': 2, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:52,949] Trial 11 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 85, 'max_depth': 7, 'min_samples_split': 2, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:53,145] Trial 12 finished with value: 0.7690875232774674 and parameters: {'classifier': 'RandomForest', 'n_estimators': 54, 'max_depth': 10, 'min_samples_split': 2, 'min_samples_leaf': 4, 'bootstrap': False}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:53,586] Trial 13 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 131, 'max_depth': 6, 'min_samples_split': 4, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 7 with value: 0.7783985102420856.
[I 2025-06-12 23:27:53,965] Trial 14 finished with value: 0.7821229050279329 and parameters: {'classifier': 'RandomForest', 'n_estimators': 114, 'max_depth': 13, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 14 with value: 0.7821229050279329.
[I 2025-06-12 23:27:54,509] Trial 15 finished with value: 0.7728119180633147 and parameters: {'classifier': 'RandomForest', 'n_estimators': 123, 'max_depth': 15, 'min_samples_split': 10, 'min_samples_leaf': 6, 'bootstrap': True}. Best is trial 14 with value: 0.7821229050279329.
[I 2025-06-12 23:27:55,219] Trial 16 finished with value: 0.7877094972067038 and parameters: {'classifier': 'RandomForest', 'n_estimators': 197, 'max_depth': 13, 'min_samples_split': 10, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:55,974] Trial 17 finished with value: 0.7802607076350093 and parameters: {'classifier': 'RandomForest', 'n_estimators': 224, 'max_depth': 14, 'min_samples_split': 10, 'min_samples_leaf': 5, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:56,700] Trial 18 finished with value: 0.7690875232774674 and parameters: {'classifier': 'RandomForest', 'n_estimators': 191, 'max_depth': 12, 'min_samples_split': 9, 'min_samples_leaf': 3, 'bootstrap': True}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:58,670] Trial 19 finished with value: 0.7486033519553073 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 192, 'learning_rate': 0.010492175364535087, 'max_depth': 13, 'min_samples_split': 9, 'min_samples_leaf': 6}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:59,089] Trial 20 finished with value: 0.7802607076350094 and parameters: {'classifier': 'RandomForest', 'n_estimators': 87, 'max_depth': 17, 'min_samples_split': 9, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:59,427] Trial 21 finished with value: 0.7802607076350093 and parameters: {'classifier': 'RandomForest', 'n_estimators': 98, 'max_depth': 17, 'min_samples_split': 10, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:27:59,785] Trial 22 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 99, 'max_depth': 17, 'min_samples_split': 9, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:28:00,233] Trial 23 finished with value: 0.7783985102420857 and parameters: {'classifier': 'RandomForest', 'n_estimators': 117, 'max_depth': 18, 'min_samples_split': 8, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:28:00,884] Trial 24 finished with value: 0.7839851024208565 and parameters: {'classifier': 'RandomForest', 'n_estimators': 177, 'max_depth': 14, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:28:01,473] Trial 25 finished with value: 0.7783985102420857 and parameters: {'classifier': 'RandomForest', 'n_estimators': 172, 'max_depth': 15, 'min_samples_split': 8, 'min_samples_leaf': 5, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:28:02,283] Trial 26 finished with value: 0.7597765363128491 and parameters: {'classifier': 'RandomForest', 'n_estimators': 206, 'max_depth': 18, 'min_samples_split': 9, 'min_samples_leaf': 4, 'bootstrap': True}. Best is trial 16 with value: 0.7877094972067038.
[I 2025-06-12 23:28:02,803] Trial 27 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 152, 'max_depth': 11, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 16 with value: 0.7877094972067039.
[I 2025-06-12 23:28:04,183] Trial 28 finished with value: 0.7448789571694601 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 147, 'learning_rate': 0.01103203667971945, 'max_depth': 11, 'min_samples_split': 7, 'min_samples_leaf': 7}. Best is trial 27 with value: 0.7877094972067039.

[I 2025-06-12 23:28:04,609] Trial 29 finished with value: 0.7523277467411545 and parameters: {'classifier': 'SVM', 'C': 9.81503189129584, 'kernel': 'poly', 'gamma': 'scale'}. Best is trial 27 with value: 0.7877094972067039.

[I 2025-06-12 23:28:05,415] Trial 30 finished with value: 0.7914338919925513 and parameters: {'classifier': 'RandomForest', 'n_estimators': 229, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 30 with value: 0.7914338919925513.

[I 2025-06-12 23:28:06,142] Trial 31 finished with value: 0.7951582867783986 and parameters: {'classifier': 'RandomForest', 'n_estimators': 227, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:06,951] Trial 32 finished with value: 0.7858472998137803 and parameters: {'classifier': 'RandomForest', 'n_estimators': 239, 'max_depth': 9, 'min_samples_split': 8, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:07,615] Trial 33 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 216, 'max_depth': 8, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:08,411] Trial 34 finished with value: 0.7765363128491621 and parameters: {'classifier': 'RandomForest', 'n_estimators': 252, 'max_depth': 7, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:08,983] Trial 35 finished with value: 0.7411545623836128 and parameters: {'classifier': 'RandomForest', 'n_estimators': 216, 'max_depth': 3, 'min_samples_split': 8, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:10,698] Trial 36 finished with value: 0.7672253258845437 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 258, 'learning_rate': 0.039556363346155726, 'max_depth': 5, 'min_samples_split': 6, 'min_samples_leaf': 2}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:10,721] Trial 37 finished with value: 0.7523277467411545 and parameters: {'classifier': 'SVM', 'C': 0.10092422231000146, 'kernel': 'poly', 'gamma': 'scale'}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:11,577] Trial 38 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 240, 'max_depth': 8, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:14,549] Trial 39 finished with value: 0.750465549348231 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 270, 'learning_rate': 0.028438486941961372, 'max_depth': 11, 'min_samples_split': 9, 'min_samples_leaf': 4}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:15,661] Trial 40 finished with value: 0.7709497206703911 and parameters: {'classifier': 'RandomForest', 'n_estimators': 292, 'max_depth': 9, 'min_samples_split': 7, 'min_samples_leaf': 2, 'bootstrap': True}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:16,821] Trial 41 finished with value: 0.7914338919925513 and parameters: {'classifier': 'RandomForest', 'n_estimators': 204, 'max_depth': 12, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:17,675] Trial 42 finished with value: 0.7914338919925513 and parameters: {'classifier': 'RandomForest', 'n_estimators': 213, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:18,319] Trial 43 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 175, 'max_depth': 10, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:18,353] Trial 44 finished with value: 0.4636871508379888 and parameters: {'classifier': 'SVM', 'C': 2.4940684206204367, 'kernel': 'sigmoid', 'gamma': 'scale'}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:19,366] Trial 45 finished with value: 0.7895716945996275 and parameters: {'classifier': 'RandomForest', 'n_estimators': 235, 'max_depth': 12, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:20,196] Trial 46 finished with value: 0.7709497206703911 and parameters: {'classifier': 'RandomForest', 'n_estimators': 241, 'max_depth': 12, 'min_samples_split': 8, 'min_samples_leaf': 9, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:21,218] Trial 47 finished with value: 0.7783985102420857 and parameters: {'classifier': 'RandomForest', 'n_estimators': 264, 'max_depth': 10, 'min_samples_split': 5, 'min_samples_leaf': 1, 'bootstrap': True}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:21,988] Trial 48 finished with value: 0.7932960893854749 and parameters: {'classifier': 'RandomForest', 'n_estimators': 226, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:22,022] Trial 49 finished with value: 0.7541899441340782 and parameters: {'classifier': 'SVM', 'C': 2.504618085195606, 'kernel': 'rbf', 'gamma': 'scale'}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:22,709] Trial 50 finished with value: 0.7858472998137803 and parameters: {'classifier': 'RandomForest', 'n_estimators': 208, 'max_depth': 8, 'min_samples_split': 7, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:23,443] Trial 51 finished with value: 0.7802607076350094 and parameters: {'classifier': 'RandomForest', 'n_estimators': 230, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:24,335] Trial 52 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 279, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:25,114] Trial 53 finished with value: 0.7821229050279329 and parameters: {'classifier': 'RandomForest', 'n_estimators': 228, 'max_depth': 12, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:25,754] Trial 54 finished with value: 0.7746741154562384 and parameters: {'classifier': 'RandomForest', 'n_estimators': 204, 'max_depth': 6, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:26,534] Trial 55 finished with value: 0.7728119180633147 and parameters: {'classifier': 'RandomForest', 'n_estimators': 244, 'max_depth': 9, 'min_samples_split': 8, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:27,268] Trial 56 finished with value: 0.7802607076350093 and parameters: {'classifier': 'RandomForest', 'n_estimators': 220, 'max_depth': 11, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:28,703] Trial 57 finished with value: 0.7560521415270017 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 188, 'learning_rate': 0.2575141429043726, 'max_depth': 10, 'min_samples_split': 9, 'min_samples_leaf': 3}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:29,771] Trial 58 finished with value: 0.7821229050279331 and parameters: {'classifier': 'RandomForest'}

t', 'n_estimators': 231, 'max_depth': 13, 'min_samples_split': 5, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:30,616] Trial 59 finished with value: 0.7746741154562384 and parameters: {'classifier': 'RandomForest', 'n_estimators': 211, 'max_depth': 7, 'min_samples_split': 3, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:31,579] Trial 60 finished with value: 0.7821229050279329 and parameters: {'classifier': 'RandomForest', 'n_estimators': 252, 'max_depth': 12, 'min_samples_split': 10, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:32,324] Trial 61 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 147, 'max_depth': 11, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:33,085] Trial 62 finished with value: 0.7895716945996276 and parameters: {'classifier': 'RandomForest', 'n_estimators': 160, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:33,686] Trial 63 finished with value: 0.7765363128491619 and parameters: {'classifier': 'RandomForest', 'n_estimators': 161, 'max_depth': 8, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:34,372] Trial 64 finished with value: 0.7914338919925513 and parameters: {'classifier': 'RandomForest', 'n_estimators': 199, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:34,998] Trial 65 finished with value: 0.7858472998137803 and parameters: {'classifier': 'RandomForest', 'n_estimators': 187, 'max_depth': 10, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:35,775] Trial 66 finished with value: 0.7672253258845437 and parameters: {'classifier': 'RandomForest', 'n_estimators': 199, 'max_depth': 9, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': True}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:36,255] Trial 67 finished with value: 0.7653631284916201 and parameters: {'classifier': 'RandomForest', 'n_estimators': 164, 'max_depth': 7, 'min_samples_split': 10, 'min_samples_leaf': 10, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:36,842] Trial 68 finished with value: 0.7858472998137803 and parameters: {'classifier': 'RandomForest', 'n_estimators': 182, 'max_depth': 8, 'min_samples_split': 9, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:36,873] Trial 69 finished with value: 0.6499068901303539 and parameters: {'classifier': 'SVM', 'C': 6.2, 'gamma': 0.001, 'kernel': 'sigmoid', 'gamma': 'auto'}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:38,418] Trial 70 finished with value: 0.7672253258845437 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 136, 'learning_rate': 0.09933206023571682, 'max_depth': 9, 'min_samples_split': 8, 'min_samples_leaf': 1}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:39,158] Trial 71 finished with value: 0.7821229050279329 and parameters: {'classifier': 'RandomForest', 'n_estimators': 225, 'max_depth': 13, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:39,807] Trial 72 finished with value: 0.7914338919925511 and parameters: {'classifier': 'RandomForest', 'n_estimators': 197, 'max_depth': 11, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:40,526] Trial 73 finished with value: 0.7895716945996276 and parameters: {'classifier': 'RandomForest', 'n_estimators': 198, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:41,211] Trial 74 finished with value: 0.7877094972067038 and parameters: {'classifier': 'RandomForest', 'n_estimators': 214, 'max_depth': 11, 'min_samples_split': 9, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:41,764] Trial 75 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 168, 'max_depth': 14, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:42,379] Trial 76 finished with value: 0.7690875232774674 and parameters: {'classifier': 'RandomForest', 'n_estimators': 201, 'max_depth': 11, 'min_samples_split': 9, 'min_samples_leaf': 7, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:43,077] Trial 77 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 222, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:43,785] Trial 78 finished with value: 0.7951582867783985 and parameters: {'classifier': 'RandomForest', 'n_estimators': 211, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:44,512] Trial 79 finished with value: 0.7616387337057727 and parameters: {'classifier': 'RandomForest', 'n_estimators': 193, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 5, 'bootstrap': True}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:45,200] Trial 80 finished with value: 0.7746741154562384 and parameters: {'classifier': 'RandomForest', 'n_estimators': 208, 'max_depth': 6, 'min_samples_split': 8, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:46,044] Trial 81 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 219, 'max_depth': 8, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:46,780] Trial 82 finished with value: 0.7858472998137803 and parameters: {'classifier': 'RandomForest', 'n_estimators': 181, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:47,486] Trial 83 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 212, 'max_depth': 11, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:28:48,106] Trial 84 finished with value: 0.7877094972067039 and parameters: {'classifier': 'RandomForest', 'n_estimators': 193, 'max_depth': 9, 'min_samples_split': 10, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:29:17,305] Trial 85 finished with value: 0.7746741154562384 and parameters: {'classifier': 'SVM', 'C': 1.2, 'gamma': 0.001, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 31 with value: 0.7951582867783986.

[I 2025-06-12 23:29:18,058] Trial 86 finished with value: 0.7877094972067038 and parameters: {'classifier': 'RandomForest', 'n_estimators': 245, 'max_depth': 8, 'min_samples_split': 10, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.


```
[I 2025-06-12 23:29:18,829] Trial 87 finished with value: 0.7802607076350093 and parameters: {'classifier': 'RandomForest', 'n_estimators': 236, 'max_depth': 12, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:21,084] Trial 88 finished with value: 0.756052141527002 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 202, 'learning_rate': 0.018953365779872427, 'max_depth': 10, 'min_samples_split': 10, 'min_samples_leaf': 2}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:21,779] Trial 89 finished with value: 0.7932960893854749 and parameters: {'classifier': 'RandomForest', 'n_estimators': 183, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:22,341] Trial 90 finished with value: 0.7821229050279329 and parameters: {'classifier': 'RandomForest', 'n_estimators': 181, 'max_depth': 9, 'min_samples_split': 8, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:22,857] Trial 91 finished with value: 0.7951582867783985 and parameters: {'classifier': 'RandomForest', 'n_estimators': 155, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:23,468] Trial 92 finished with value: 0.7765363128491621 and parameters: {'classifier': 'RandomForest', 'n_estimators': 170, 'max_depth': 9, 'min_samples_split': 6, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:23,964] Trial 93 finished with value: 0.7783985102420856 and parameters: {'classifier': 'RandomForest', 'n_estimators': 154, 'max_depth': 8, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:24,687] Trial 94 finished with value: 0.7802607076350093 and parameters: {'classifier': 'RandomForest', 'n_estimators': 226, 'max_depth': 7, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:25,382] Trial 95 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 207, 'max_depth': 10, 'min_samples_split': 8, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:26,200] Trial 96 finished with value: 0.7709497206703911 and parameters: {'classifier': 'RandomForest', 'n_estimators': 217, 'max_depth': 11, 'min_samples_split': 9, 'min_samples_leaf': 2, 'bootstrap': True}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:26,955] Trial 97 finished with value: 0.7877094972067038 and parameters: {'classifier': 'RandomForest', 'n_estimators': 234, 'max_depth': 8, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:27,188] Trial 98 finished with value: 0.7728119180633147 and parameters: {'classifier': 'RandomForest', 'n_estimators': 66, 'max_depth': 9, 'min_samples_split': 8, 'min_samples_leaf': 2, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
[I 2025-06-12 23:29:27,868] Trial 99 finished with value: 0.7951582867783985 and parameters: {'classifier': 'RandomForest', 'n_estimators': 187, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}. Best is trial 31 with value: 0.7951582867783986.
```

```
In [96]: # Retrieve the best trial
best_trial = study.best_trial
print("Best trial parameters:", best_trial.params)
print("Best trial accuracy:", best_trial.value)
```

```
Best trial parameters: {'classifier': 'RandomForest', 'n_estimators': 227, 'max_depth': 9, 'min_samples_split': 9, 'min_samples_leaf': 1, 'bootstrap': False}
Best trial accuracy: 0.7951582867783986
```

This code to see which model and hyperparameter values performed best, and what accuracy they achieved.

This helps you understand which settings you should use in your final model.

```
In [97]: study.trials_dataframe()
```

Out[97]:

	number	value	datetime_start	datetime_complete	duration	params_C	params_bootstrap	params_classifier	
0	0	0.759777	2025-06-12 23:27:39.583426	2025-06-12 23:27:40.986304	0 days 00:00:01.402878	NaN	True	RandomForest	
1	1	0.746741	2025-06-12 23:27:40.986304	2025-06-12 23:27:43.781357	0 days 00:00:02.795053	NaN	NaN	GradientBoosting	
2	2	0.772812	2025-06-12 23:27:43.783359	2025-06-12 23:27:45.235975	0 days 00:00:01.452616	0.371053	NaN	SVM	
3	3	0.769088	2025-06-12 23:27:45.236974	2025-06-12 23:27:46.073216	0 days 00:00:00.836242	NaN	False	RandomForest	
4	4	0.741155	2025-06-12 23:27:46.074228	2025-06-12 23:27:47.848181	0 days 00:00:01.773953	NaN	NaN	GradientBoosting	
...	...	...	...	...	...	...	...	...	...
95	95	0.783985	2025-06-12 23:29:24.688764	2025-06-12 23:29:25.382103	0 days 00:00:00.693339	NaN	False	RandomForest	
96	96	0.770950	2025-06-12 23:29:25.383098	2025-06-12 23:29:26.200694	0 days 00:00:00.817596	NaN	True	RandomForest	
97	97	0.787709	2025-06-12 23:29:26.201698	2025-06-12 23:29:26.955034	0 days 00:00:00.753336	NaN	False	RandomForest	
98	98	0.772812	2025-06-12 23:29:26.955034	2025-06-12 23:29:27.188559	0 days 00:00:00.233525	NaN	False	RandomForest	
99	99	0.795158	2025-06-12 23:29:27.188559	2025-06-12 23:29:27.868405	0 days 00:00:00.679846	NaN	False	RandomForest	

100 rows × 16 columns

In [98]:

study.trials_dataframe()['params_classifier'].value_counts()

Out[98]:

params_classifier
RandomForest 80
GradientBoosting 10
SVM 10
Name: count, dtype: int64

In [99]:

study.trials_dataframe().groupby('params_classifier')['value'].mean()

Out[99]:

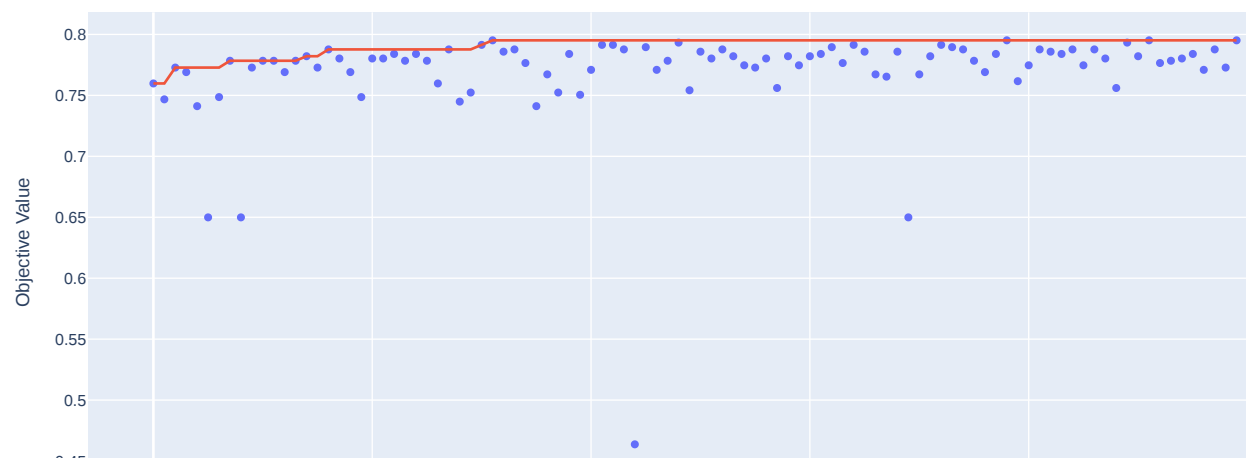
params_classifier
GradientBoosting 0.752700
RandomForest 0.780796
SVM 0.699255
Name: value, dtype: float64

In []:

In [100...]

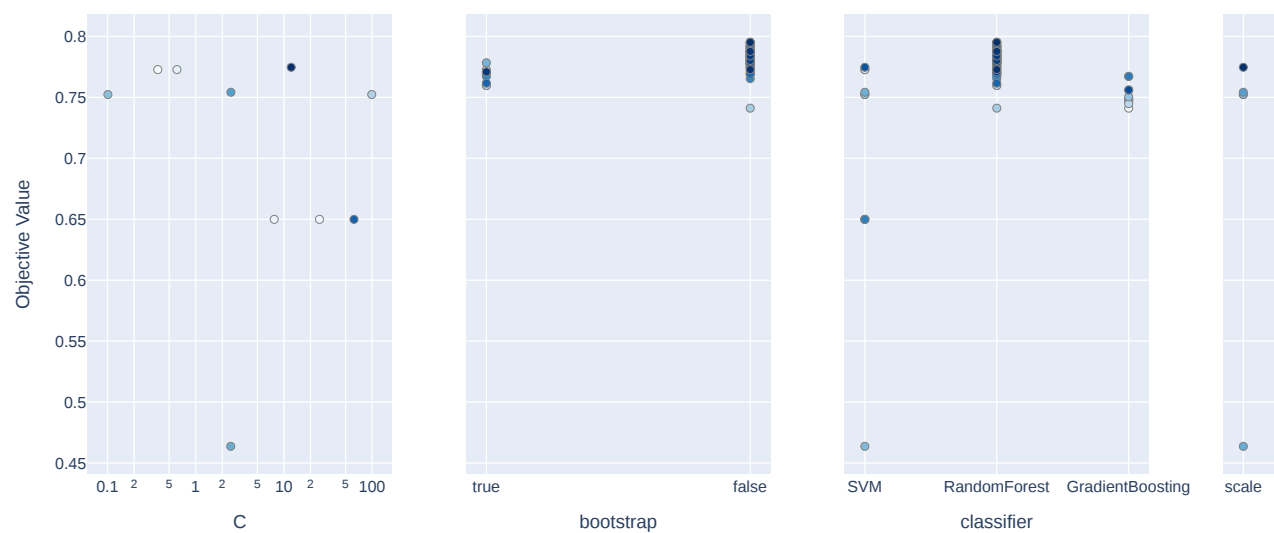
1. Optimization History
plot_optimization_history(study).show()

Optimization History Plot



```
In [101... plot_slice(study).show()
```

Slice Plot



```
In [102... plot_param_importances(study).show()
```

Hyperparameter Importances



In []:

In this project, we used Optuna to perform hyperparameter tuning on multiple machine learning models—Support Vector Machine (SVM), Random Forest, and Gradient Boosting—to find the best-performing model and its optimal settings. We defined an objective function that allowed Optuna to suggest different values for each model’s key hyperparameters, then evaluated each combination using cross-validation to measure accuracy. We used a sampler to either randomly or systematically try different settings. After running several trials, we retrieved the best trial using `study.best_trial` to identify the most accurate model along with the exact parameter values that achieved that result. This process helps us automatically optimize our machine learning models for better performance.

In []:

In []:

In []: